



The neural correlates of flow experience explored with transcranial direct current stimulation

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Abstract

The experience of flow ensues when humans engage in a demanding task while task demands are balanced with the individual's level of skill or ability. Here, we further tested the hypothesis that the medial prefrontal cortex (MPFC) plays a causal role in mediating flow experience using transcranial direct current stimulation (tDCS) to interfere with MPFC's deactivation evoked by a flow paradigm and measured by magnetic resonance (MR)-based perfusion imaging. In a balanced, within-subjects repeated measure design, three treatments of tDCS (sham, anodal, cathodal) were applied in a sample of 22 healthy male participants. tDCS-modulatory effects on flow-specific regional cerebral blood flow (rCBF) and subjective flow experience significantly depended on participants' baseline level of flow experience during sham tDCS. Those participants with lower-flow experience during sham tDCS (LF) benefitted from tDCS, particularly from the anodal polarity, whereas both active treatments did not substantially affect subjects with relatively higher baseline flow experience (HF). Functionally, in LF subjects, relative deactivation of the right amygdala got more pronounced under anodal and cathodal tDCS, and changed inconsistently in HF subjects. Inter-individual regression analyses of rCBF data suggested that involvement of the subgenual anterior cingulate cortex appears crucial for affecting the response pattern in the right amygdala and can be modulated by tDCS. Present data support the notion that valuable insights into the neural mechanism of flow can be obtained using tDCS. However, a clearer understanding of tDCS' baseline dependency in terms of individual variations in brain connectivity states appears a necessary prerequisite to exploit this technique further.

Keywords Flow experience · tDCS · Perfusion imaging · MPFC · Amygdala · sgACC

Introduction

Flow experience describes a state of effortless involvement, suspension of self-reflective thoughts, and intrinsic motivation during engagement in an activity whose demands are in optimal fit to one's level of skill or ability (Csikszentmihalyi 1975).

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Previous research investigating the neural correlates and mechanisms of flow experience utilized a mental arithmetic task with varying levels of difficulty to create a “flow” (F) condition and control conditions of “boredom” (B) and “overload” (O) (Ulrich et al. 2014, 2016b). In the F condition, difficulty of the task was continuously adjusted to participants' arithmetic ability prior to and evolving during the experiment. Under flow, relative to control conditions, participants gave reliably higher rating scores on Likert-scaled statements assessing joy, willingness to repeat the task, and perceived fit between demands and ability. At the neural level, particularly the amygdala and the medial prefrontal cortex (MPFC) appeared to be important mediators of the flow experience. Both structures exhibited robustly decreased activation under flow relative to control conditions, and this pattern significantly predicted the degree of subjective flow experience. With diminished self-reflection representing a core feature of the flow experience, the finding of relatively decreased MPFC activity during flow aligns

well with the MPFC's well-known role in self-referential processing (Gusnard et al. 2001; Zysset et al. 2003; Goldberg et al. 2006; D'Argembeau et al. 2007; Johnson et al. 2009; van Buuren et al. 2010; Brewer et al. 2011; Jenkins and Mitchell 2011; Whitfield-Gabrieli et al. 2011).

In the present study, we further investigated the presumably causative role of the MPFC for the experience of flow using transcranial direct current stimulation (tDCS) to interact with the MPFC's activity level while subjects performed the above-described arithmetic task. Regional cerebral blood flow (rCBF) was measured with magnetic resonance (MR)-based perfusion imaging implemented as pseudo-continuous arterial spin labeling.

Transcranial direct current stimulation is a non-invasive stimulation technique with a small electric current applied by an electrode under which activation of cortical neurons is modulated. Since the early endeavors (e.g., Creutzfeldt et al. 1962; Purpura and McMurtry 1965) in experimental animals, use of tDCS has been revived in the last two decades for in vivo application in humans (for reviews see, e.g., Paulus 2011; Krause and Cohen Kadosh 2014).

Particularly in the field of modulating excitability of the human motor cortex using motor-evoked potential (MEP) amplitudes as the main dependent variable, it has been shown that the current direction may associate with effects in opposite direction. Anodal tDCS is assumed to hyperpolarize the dendrites and to depolarize the soma, resulting in a net increase in neuronal excitability. Conversely, cathodal tDCS is thought to decrease neuronal excitability (Bikson et al. 2004; Nitsche et al. 2008). However, with more cognitive paradigms and dependent variables other than MEP amplitudes, it has also been shown that this rather simple dichotomy of polarity effects does not necessarily hold (e.g., Jacobson et al. 2012), and that individual differences (e.g., age, gender, hormones), and particularly differences in the "neural state" may determine the modulating effect by interacting with anodal or cathodal tDCS (Krause and Cohen Kadosh 2014; Hsu et al. 2016).

While some of these variables can be more easily controlled by rigorously adopting the study design, control of the initial brain state is much more difficult because excitability levels prior to the stimulation are usually not known (Krause and Cohen Kadosh 2014). As an estimator, it has been suggested that this unknown state can be predicted by individual variability in behavior or performance measures derived from the sham condition to serve as pre-existing baseline difference by which participants are split into groups of presumably different extents of cortical excitability and inhibition (Tseng et al. 2012; Hsu et al. 2014, 2016; Krause and Cohen Kadosh 2014). Indeed, baseline dependence of tDCS effects has been shown for various functional domains where tDCS effects would have been underestimated had individually different baseline performances not been considered (Turkeltaub et al. 2012; Furuya

et al. 2014; Hsu et al. 2014, 2016; Benwell et al. 2015; Learmonth et al. 2015; Gözenman and Berryhill 2016; Heinen et al. 2016; Reinhart et al. 2016; Habich et al. 2017; McConathey et al. 2017; Perceval et al. 2017; Schaal et al. 2017; Strobach et al. 2017). Interestingly, low performers were usually more responsive to tDCS, while there were no or no strong effects for high performers, ruling out simple regression to the mean in motivating this overall result pattern.

Based on previous reports about successful tDCS modulation of the MPFC (Civai et al. 2014; Manuel et al. 2014; Nakamura and Kawabata 2015; Hämmerer et al. 2016; Adenzato et al. 2017; Bertossi et al. 2017), we made the following predictions: given the frequently reported brain state dependency of tDCS effects, we did not expect to observe any effects of tDCS on subjective flow experience when considering the entire sample. However, we did expect tDCS effects after dividing the sample into two groups of subjects reporting more (higher-flow group, HF) versus less (lower-flow group, LF) subjective flow experience during sham stimulation. To prove whether this split into two groups was not merely random, and based on our previous results of the neural correlates of flow (Ulrich et al. 2014, 2016b), we also predicted that subgroups would differ in their pattern of neural activation during the flow condition under sham stimulation: the group expressing lower baseline levels of subjectively experienced flow was expected to show significantly less differential deactivation of the MPFC and amygdala compared to the subgroup with higher levels of subjectively reported flow experience. Finally, given the overall result pattern (López-Alonso et al. 2014) that "low performers" benefitted more from tDCS than "high performers", we also hypothesized that application of tDCS should increase the experience of flow in the LF group but not in the HF group. This change in subjectively experienced flow should be accompanied by greater sham-to-active tDCS decrease in MPFC and amygdala activity in the LF group compared to the HF group. Given the novelty of the entire research area, that is, concomitant tDCS during functional neuroimaging of flow experience, no clear-cut predictions could be set up regarding polarity effects of tDCS. Nonetheless, based on the common assumption (Jacobson et al. 2012) that cathodal tDCS induces a decrease of cortical excitability due to its reported soma hyperpolarizing effect, a greater degree of cortical deactivation during flow was more likely to be expected under cathodal tDCS than under anodal tDCS.

Materials and methods

Participants

Twenty-two male, right-handed students recruited from the local university, with an average age of 24.9 years (standard deviation, SD: 2.2 years), participated in the study. None of

the participants reported any psychiatric/neurological disorders or contraindications with regard to MRI or tDCS. The study was in accordance with the Declaration of Helsinki, and approved by the Ethics Committee at Ulm University. Written informed consent was obtained prior to the experiment. Financial compensation was 60 EUR.

Experimental design

The study was implemented in a within-subject design. Each participant appeared for 3 days in our laboratory to receive either sham tDCS, anodal tDCS, or cathodal tDCS. Treatment application order was fully balanced across the subjects. The experimental paradigm was always the same: participants performed several blocks of a mental arithmetic task, established in our previous work (Ulrich et al. 2014, 2016b). A block lasted about 170 s and consisted of a varying number of math tasks which required the addition of two or more numbers, presented serially with short intermittent breaks. The result was to be entered using an on-screen keyboard. Upon submitting the result, or after a timeout of 18 s, a 4 s break followed during which the meaningless string “xxx + x” appeared on the screen. The next math expression followed thereafter. Each block was assigned one of three possible levels of task difficulty: boredom, flow, or overload. In the boredom condition, a number between 1 and 9 had to be added to a number between 100 and 109. A computer algorithm ensured that the sum was always equal to or smaller than 110. Blocks of the flow condition started at a difficulty level that was in balance with participants’ calculation ability, estimated during a preparation phase (see below). Afterwards, task difficulty was continually and automatically adjusted according to participants’ responses: if participants entered two succinct results correctly, task difficulty increased by one level, and if both results were incorrect, difficulty decreased by one level. A level increased either by replacing the last summand by a two-digit number (in case it had been a single-digit number), or by amending the math expression by an additional one-digit summand. The two steps were reversed in cases of level decrease. In the overload condition, the starting level of task difficulty was set three levels higher than that of the flow condition, and the difficulty was maintained equal to or higher than the starting level throughout the block. Subjects did not receive explicit feedback about response accuracy in any of the conditions.

Per treatment session, there were three blocks of each task condition, alternating with rest blocks of 20 s duration during which subjects had to fixate a black cross centered on a white screen. Perfusion imaging was only performed during task blocks. Blocks were ordered according to the following two predefined sequences: –B–F–O–F–O–B–O–B–F–, and –B–O–F–O–F–B–F–B–O– (rest blocks are reflected by the hyphen). While both orders were counterbalanced across

participants, within each subject a given order was maintained over the three treatment sessions.

Immediately after completion of each task block, the following three 7-point Likert-scaled statements were presented to estimate the degree of experienced flow: “I would love to solve math calculations of that kind again”, “Task demands were well matched to my ability”, and “I was thrilled”. Responses could range from “I do not agree at all” to “I completely agree”. Mean duration per question period (comprising the three items) was 10.1 s (SD 2.2 s) under sham tDCS, 9.9 s (SD 2.1 s) under anodal tDCS, and 10.1 s (SD 2.1 s) under cathodal tDCS ($F(2, 46) = 0.15$, $p = 0.863$). Altogether, the duration of the experiment was about 30 min per treatment session (task blocks: 25.5 min, rest blocks: 3.3 min, question periods: 1.5 min).

In a preparation phase prior to scanning, participants underwent two practice blocks. The first block consisted of the B condition (3 min) to familiarize participants with the trackball (NAtA TECHNOLOGIES, Coquitlam, Canada). The second block featured the flow condition, extended to 5 min and starting at level 1. Participants’ individual math ability was estimated by averaging the difficulty level achieved during the last 25% of presented math expressions. The resulting mean level was used as “starting level”. It was determined for each individual treatment session.

Numbers and characters were presented in black font on a white background, centered on a 32” LCD display (NordicNeuroLab AS, Bergen, Norway) at 1280 × 720 pixels resolution, projected to participants’ eyes by a mirror. The task was programmed in Scala (v2.9.01) running in Java Runtime Environment (v6.0.05) under 64 bit Windows 7, Service Pack 1 on a standard PC (Core i7-4790 CPU @ 3.60 GHz; 16 GB RAM).

Transcranial direct current stimulation

While participants performed the experimental task, one of three tDCS treatments was applied, counterbalanced across subjects: sham tDCS, anodal tDCS, or cathodal tDCS. Treatments were administered on different days separated by at least 72 h (mean delay between first and second visit: 6.5 days, SD: 3.1 days, maximum: 14 days; mean delay between second and third visit: 7.9 days, SD: 5.1 days, maximum: 18 days).

Stimulation was delivered using the device “DC-Stimulator MR” (neuroConn GmbH, Ilmenau, Germany), combined with MRI compatible rubber electrodes (size: 7 cm × 5 cm) and Ten20 Conductive Paste (Weaver and Company, Aurora, CO, USA). To stimulate the medial prefrontal cortex, the active electrode was placed horizontally over position Fpz, whereas the extracephalic reference electrode was attached above the right shoulder. Both electrodes were kept in place with elastic bands. For all treatments, the final current

intensity was 1.5 mA, resulting in a current density of about 0.043 mA/cm². Currents were faded in and out for 10 s each. Sham tDCS consisted of brief stimulation for 30 s duration (excluding fading). Anodal and cathodal tDCS lasted 30 min and otherwise differed in current direction only. TDCS was initiated roughly 30 s before the experiment.

MRI data acquisition

Magnetic resonance imaging data were acquired on a 3 T MAGNETOM Prisma (Siemens AG, Erlangen, Germany) with a 20 channel head/neck coil. Perfusion imaging was performed using pseudo-continuous arterial spin labeling (pCASL) implemented as a 3D gradient echo spin echo (GRASE) sequence, as described in Boland et al. (2018). The labeling plane was placed about 85 mm inferior to the bi-commissural line, perpendicular to the carotids. In accordance with the recommendations of a recent white paper (Alsop et al. 2015), the following parameters were used: repetition time (TR) = 3900 ms (2 TRs per volume), echo time (TE) = 20 ms, field of view = 224 mm, matrix size = 64 × 64, slices per slab = 40, “center-out” view ordering along second phase encoding axis, slice orientation: transversal, acquisition: parallel to the bi-commissural line, excitation/refocusing flip angle = 90°/120° (slab-selective), PAT factor = 2 (GRAPPA, along second phase encoding direction), bandwidth = 2298 Hz/pixel, bolus length = 2400 ms, postlabeling delay = 1000 ms, pre-saturation and background suppression with 2 inversion pulses, voxel size = 3.5 mm × 3.5 mm × 3.5 mm. One perfusion block comprised acquisitions of 9 labeled and 9 control images, and two M0 images prior to perfusion scanning such that entire duration lasted around 170 s.

A high resolution T1-weighted structural image was obtained by administering a 3D magnetization prepared rapid acquisition gradient echo sequence (TR = 2300 ms, TE = 2.61 ms, inversion time = 1100 ms, flip angle = 9°, field of view = 256 mm, matrix size = 256 × 256, voxel volume = 1 mm³, slice orientation: sagittal; PAT factor = 2 (GRAPPA mode); scan time = 303 s).

Assessment of flow experience during the experiment

Like in our previous fMRI studies investigating flow (Ulrich et al. 2014, 2016b), we utilized the “flow index” as the pivotal behavioral measure of subjective flow experiences. The flow index was computed from the rating scores given on the three Likert-scaled items after each experimental task block. First, for each treatment session, ratings were averaged item-wise across the three repeated blocks per condition. Then, the average scores were differentially weighted (“−B + 2F − O”) per item, and the outcome was summed

up across items. This was done for each treatment session, hence providing three flow indices per participant.

Post-scan questionnaires

After each scanning session, participants were administered a German version of the Swedish Flow Proneness Questionnaire (SFPQ; Ullén et al. 2012) to evaluate the individual proneness to experience flow in three everyday activities: work, maintenance, and leisure. There were five possible responses: “never”, “rarely”, “sometimes”, “often”, and “everyday, or almost everyday”. In the domain “work”, the statement “When you do something at work, how often does it happen that ...” was replaced by “When you study for exams, prepare a presentation, or write a research paper, how often does it happen that ...” to account for the fact that all participants were students.

We also assessed participants’ preferences for mental arithmetic and for reading (as a control) by asking: “How much do you like performing mental arithmetic?”, and “How much do you like reading?”. Responses were given on visual analog scales ranging from “not at all” to “as much as possible”.

Side effects of tDCS were recorded using the questionnaire introduced by Fertonani et al. (2010).

Data analysis

Flow index

The flow indices per subject and treatment were passed to a repeated measures one-way analysis of variance (ANOVA) to test for significant differences in mean flow experience between tDCS treatments.

As expected from previous literature (see “Introduction”) and reported in greater details further below, there was no significant effect of treatment on subjectively experienced flow levels (see Fig. 1a). Then, to explore whether tDCS modulation effects were dependent on the baseline level of flow experienced during sham tDCS, we divided participants into two subgroups according to a median split on the flow index obtained from the sham tDCS session (see Fig. 1b).

These two subsamples will be referred to as “lower-flow” (LF) and “higher-flow” (HF) groups in the remaining text, and a between-group factor was added to the repeated measures ANOVAs on the flow index to test for significant interactions between factors group (LF vs. HF) and treatment (sham tDCS vs. anodal tDCS; sham tDCS vs. cathodal tDCS). As a consequence, the median split-based subdivision of our sample was incorporated into all further steps of data analysis.

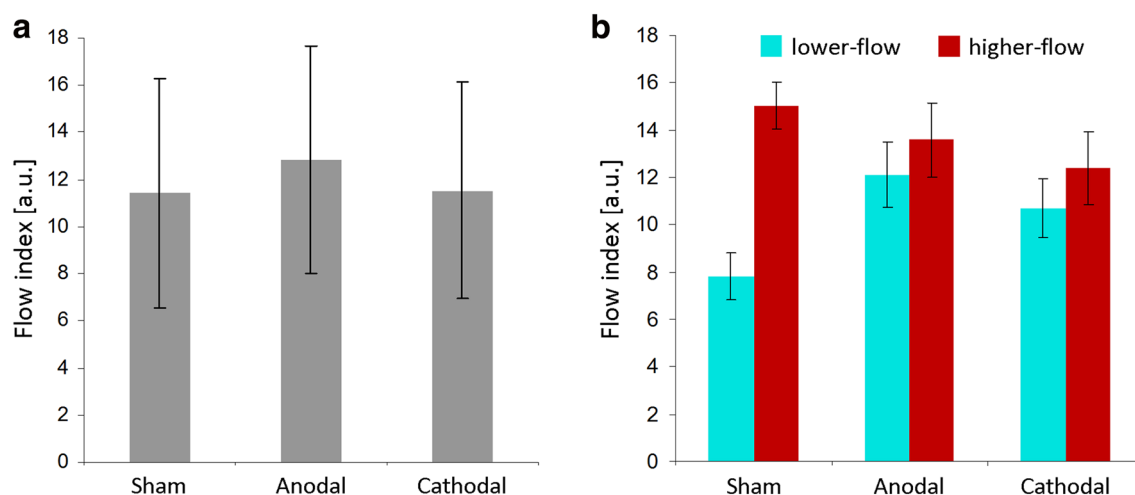


Fig. 1 **a** Flow index under sham, anodal, and cathodal tDCS ($n=22$); **b** Flow index under sham, anodal, and cathodal tDCS for the LF (lower-flow experience) and HF (higher-flow experience) groups. Error bars denote the standard error of the mean ($n=11$ per group). *a.u.* arbitrary unit

Task performance and post-scan questionnaires

Behavioral data were tested with two aims: first, we were interested whether the distinction between LF and HF participants might have also been reflected by other behavioral measures (as confounding moderator variables). For that reason, two-sample t tests were applied to the following data: (1) task performance (percentage of correctly solved math calculations) associated with the flow condition under sham tDCS; (2) total scores on the SFPQ; (3) preferences for mental arithmetic and reading; (4) tDCS side effects [Fertonani et al. (2010) questionnaire]. Our second aim was to test for possible interactions between factors group (LF, HF), treatment (sham, anodal, cathodal tDCS), and—where applicable—condition (B, F, O). Hence, repeated measures ANOVAs were performed on task performance (including all conditions, B, F, and O), on the SFPQ total score, and on the individual Fertonani items. Results were assessed at a significance level of $p < 0.050$.

MRI data

Imaging data preprocessing and statistical analyses were performed with MATLAB 2013b (MathWorks Inc., Natick, MA, USA) and Statistical Parametric Mapping (SPM12 r6685, Wellcome Department of Cognitive Neurology, London, UK). Labeled and control images were separately realigned and coregistered block-wise. Afterwards, the difference between labeled and control images was computed and converted into absolute rCBF images (unit: ml/100 g tissue/min) which were then coregistered across all 27 blocks (3 treatments $\times$ 3 conditions $\times$ 3 blocks per condition). For normalization, the mean rCBF image of block 1 served as reference image to which the T1 image was coregistered.

The T1 image was segmented and the resulting forward deformations were used to spatially normalize all images to the Montreal Neurological Institute (MNI) space. The mean rCBF images were smoothed using a Gaussian kernel with 10 mm full width at half maximum. After preprocessing, all images had a voxel size of 2 mm $\times$ 2 mm $\times$ 2 mm.

A general linear model was used for voxel-wise averaging of the rCBF data for each subject. The model included the 27 regressors of interest (one regressor per block, see above) plus the volume mean over time as a covariate to reduce the effect of spatially coherent noise (Wang et al. 2003). Upon model estimation, one-sided t -contrasts were created to represent average rCBF across the three blocks of each treatment-by-condition combination. Those nine contrast images (B, F, and O during sham, anodal, and cathodal tDCS) from each subject were submitted to a random-effects analysis, implemented as flexible factorial design, with factors group, treatment-by-condition, and subject (to remove subjects' related variability).

After model estimation, several hypotheses were investigated using one-sided t -contrasts. Suggested by our previous findings (Ulrich et al. 2014, 2016b), the medial prefrontal cortex and the amygdala were seen as the most relevant brain regions for the emergence of flow experience. Therefore, we restricted statistical tests to the MPFC and bilateral amygdala as regions of interest (ROI), defined using the Human Brainnetome Atlas (Fan et al. 2016). Our first aim was to test whether the flow index-derived subdivision of the $n=22$ sample into LF and HF groups was reflected at the neural level. From Fig. 1b, it is evident that under sham tDCS both groups had a significant experience of flow (see also Results), although to different degrees. Therefore, we tested for common, significant U-shaped activity between groups using the conjunction of the contrasts

[Sham_B – 2*Sham_F + Sham_O] in both groups. Second, Fig. 1b shows that the HF group experienced flow to a significantly greater degree than the LF group. Accordingly, we expected to find a more pronounced U-shaped brain activation pattern for HF compared with LF participants, which was tested for significance using the same contrast as above for between-group comparison [(Sham_B_{HF} – 2*Sham_F_{HF} + Sham_O_{HF}) – (Sham_B_{LF} – 2*Sham_F_{LF} + Sham_O_{LF})]. Next, we tested for group-dependent, tDCS-induced modulation of brain activity. Again suggested by Fig. 1b, LF subjects experienced more flow during anodal and cathodal tDCS relative to sham tDCS. This directional group-by-treatment interaction was addressed at the neural level in two separate comparisons, “anodal tDCS vs. sham”, and “cathodal tDCS vs. sham”, for which the following interaction contrasts were formulated: [(Anodal_U_{LF} – Sham_U_{LF}) – (Anodal_U_{HF} – Sham_U_{HF})], and [(Cathodal_U_{LF} – Sham_U_{LF}) – (Cathodal_U_{HF} – Sham_U_{HF})], with “U” reflecting the contrast “B – 2*F + O”. Accounting for the ROI approach and the lower contrast-to-noise ratio associated with perfusion MRI relative to BOLD imaging (Yang et al. 2005; Liu and Brown 2007; Viviani et al. 2011; Wang et al. 2011; Detre et al. 2012; Donahue and Jezard 2013), significance was always assessed at a voxel height threshold of $p < 0.050$ and a cluster extent threshold of $k = 30$ contiguous voxels.

Results

Flow index

As expected, the flow index did not significantly differ between sham tDCS, anodal tDCS, and cathodal tDCS when tested across the whole sample of 22 participants ($F(2, 42) = 1.08$, $p = 0.347$; see Fig. 1a). However, as illustrated in Fig. 1b, after using a median split on the sham tDCS-derived flow index, there were significant interactions of factors group and treatment in both ANOVAs comparing sham with anodal tDCS ($F(1, 20) = 8.40$, $p = 0.009$), and sham with cathodal tDCS ($F(1, 20) = 6.43$, $p = 0.020$), respectively (see also Online Resource 1). While the sham flow index significantly differed from zero in either group

(LF: $t(10) = 8.03$, $p < 0.001$; HF: $t(10) = 15.09$, $p < 0.001$), post hoc testing (Newman–Keuls tests) revealed that the between-group difference under sham stimulation was significant ($p < 0.002$). Further post hoc testing also revealed that the increase in flow experience in the LF group was significant under anodal ($p = 0.006$), but not under cathodal ($p = 0.076$) tDCS when either tDCS was compared against sham. The numerical decrease in the HF group was not significant, neither under anodal tDCS ($p = 0.313$) nor under cathodal tDCS ($p = 0.102$).

Task performance and questionnaire data

Mean task performance is listed in Table 1. Performance in the flow condition under sham tDCS did not significantly differ between both groups ($t(20) = 1.42$, $p = 0.172$). The results from the ANOVA revealed a significant main effect of condition ($F(2, 40) = 537.03$, $p < 0.001$), reflecting a linear decrease in correctly solved math calculations from 99.1% for boredom, 60.3% for flow, and 24.3% for overload (averaged over groups and treatments). Other main effects or interactions were not significant (all $p > 0.100$).

Mean total scores on the SFPQ reflecting subjects' proneness to experience flow were of comparable magnitude for both groups (LF: 3.7, SD: 0.3; HF: 3.5, SD: 0.4), and the slight numerical difference did not reach significance ($t(20) = 1.89$, $p = 0.074$).

Preferences for math and reading did also not significantly differ between groups: mean preference for math: LF: 5.1 (SD: 2.9), HF: 6.2 (SD: 2.1), $t(20) = 1.08$, $p = 0.293$. Mean preference for reading: LF: 8.2 (SD: 1.1), HF: 7.3 (SD: 2.2), $t(20) = 1.17$, $p = 0.257$.

Reported side effects of tDCS (Fertonani et al. 2010) are summarized in Table 2. Under sham tDCS, there were no significant group differences in any of the individual questionnaire items such as itchiness or pain (all $t(20) < 1.30$, all $p > 0.211$). Item-wise ANOVAs testing for possible interactions between factors group and treatment were also not significant (all $F(2, 40) < 0.93$, all $p > 0.406$).

In summary, none of the behavioral measures showed significant group differences under sham tDCS or significant interactions between factors group and treatment.

Table 1 Mean task performance as percent of correctly solved math calculations per treatment and condition

	Sham tDCS			Anodal tDCS			Cathodal tDCS		
	Boredom	Flow	Overload	Boredom	Flow	Overload	Boredom	Flow	Overload
Lower-flow group	99.3 (1.3)	53.8 (10.5)	21.0 (20.8)	99.0 (1.4)	61.7 (11.0)	21.1 (15.5)	98.9 (1.9)	61.4 (7.5)	22.9 (18.9)
Higher-flow group	99.5 (0.8)	63.4 (11.8)	24.2 (17.9)	99.0 (2.1)	60.8 (15.5)	31.5 (19.2)	98.8 (2.0)	60.7 (7.2)	25.1 (15.4)

Values in parentheses denote standard deviations

Perfusion data

Sham stimulation (baseline)

Before starting the inference analysis of tDCS treatment effects, we first made use of the perfusion data to test whether splitting the entire sample into two groups along participants' subjective flow experience (i.e., flow index) would also be reflected in the neural data. A conjunction analysis revealed a conjoint U-shaped effect in the dorsal MPFC under sham tDCS across both groups (see Fig. 2a; Table 3) at the same location as observed in our previous studies, but not in the amygdala, suggesting that the U-shaped response pattern in that region was not of the same magnitude in both groups. This was tested by contrasting the U-shaped response pattern between groups, which was indeed significantly more pronounced in the right amygdala for the HF group compared with the LF group (see Fig. 2b; Table 3), supporting the observance of a greater subjective flow index of the HF group.

tDCS treatment effects

To infer significant treatment effects, this factor was next added to the group-by-condition interactions above. When comparing sham tDCS with anodal tDCS, this three-way interaction revealed a significant effect in the right amygdala where the U-shaped response pattern increased from sham to anodal tDCS in the LF group and decreased in the HF group (see Fig. 3a; Table 3, and Online Resource 1; please note that a decrease in the figure refers to an increase in U-shaped response pattern). A similar pattern was observed for cathodal tDCS (see Fig. 3b; Table 3), although at a reduced extent of voxels bearing the significant three-way interaction effect. Again, the U-shaped response pattern increased in the LF group and decreased in the HF group, but less than under anodal tDCS.

Exploratory regression analyses of rCBF data

Sham stimulation (baseline)

In a next step we explored whether relationships between the dorsal MPFC and the right amygdala might explain the state-dependent subdivision of our sample into participants with more vs. less subjective flow experience under sham stimulation. To this end, per subject, we computed the magnitude of U-shaped (i.e., $B - 2F + O$) rCBF within the dorsal MPFC cluster (averaged across voxels) where both groups had shown a conjoint U-shaped response pattern (see Fig. 2a). These values were then entered as a regressor in SPM12 together with the contrast images representing the same signal characteristic (i.e., U-shape) as dependent variable. A group factor was also included in the model. Between-group comparisons of voxel-wise regression slopes across subjects were performed within the amygdala and the MPFC.

Between-group comparison of regression coefficients showed greater positive relationships in the HF group relative to the LF group in a portion of the MPFC adjacent to (and partly overlapping with) the cluster from which the regressor had been constructed. Notably, positive relationships were also observed in the subgenual part of the anterior cingulate cortex (sgACC) and right amygdala (for a summary, see Fig. 4a; Table 4). The inverted contrast asking for greater positive relationships between U-shaped response patterns in the LF group compared to the HF group revealed clusters in the ventral MPFC, in the dorsal MPFC, and in the left posterior rectal gyrus (see Table 4).

Anodal tDCS

In a second regression analysis we asked whether and how the pattern described above would change under anodal stimulation (we focused on anodal stimulation since the signal pattern in the right amygdala differentiating LF and HF subjects was rather the same under cathodal tDCS, see Fig. 3a, b). Applying the same rationale as above, the

Table 2 Side effects of tDCS treatments assessed by the Fertonani et al. (2010) questionnaire

	Lower-flow			Higher-flow			Sham group difference		Group-by-treatment interaction	
	Sham	Anodal	Cathodal	Sham	Anodal	Cathodal	<i>t</i> (20)	<i>p</i>	<i>F</i> (2, 40)	<i>p</i>
Itchiness	0.4 (0.7)	0.5 (0.8)	0.5 (0.9)	0.3 (0.5)	0.3 (0.5)	0.3 (0.5)	0.37	0.717	0.20	0.821
Pain	0.2 (0.4)	0.2 (0.4)	0.1 (0.3)	0.1 (0.3)	0.5 (0.8)	0.3 (0.9)	0.60	0.557	0.92	0.406
Burning	0.5 (0.7)	0.7 (0.6)	0.7 (0.6)	0.9 (0.9)	1.3 (1.3)	0.8 (0.9)	1.29	0.211	0.70	0.503
Warmth/heat	0.9 (0.9)	0.9 (0.9)	0.9 (0.8)	0.9 (0.9)	1.0 (0.9)	1.0 (0.8)	–	–	0.03	0.974
Iron taste	0.0 (0.0)	0.0 (0.0)	0.0 (0.0)	0.1 (0.3)	0.0 (0.0)	0.0 (0.0)	1.00	0.329	–	–
Fatigue	0.7 (0.6)	0.7 (1.0)	0.9 (1.0)	0.9 (1.0)	0.5 (0.5)	1.1 (1.2)	0.49	0.629	0.43	0.656

Values are means (standard deviations in parentheses)

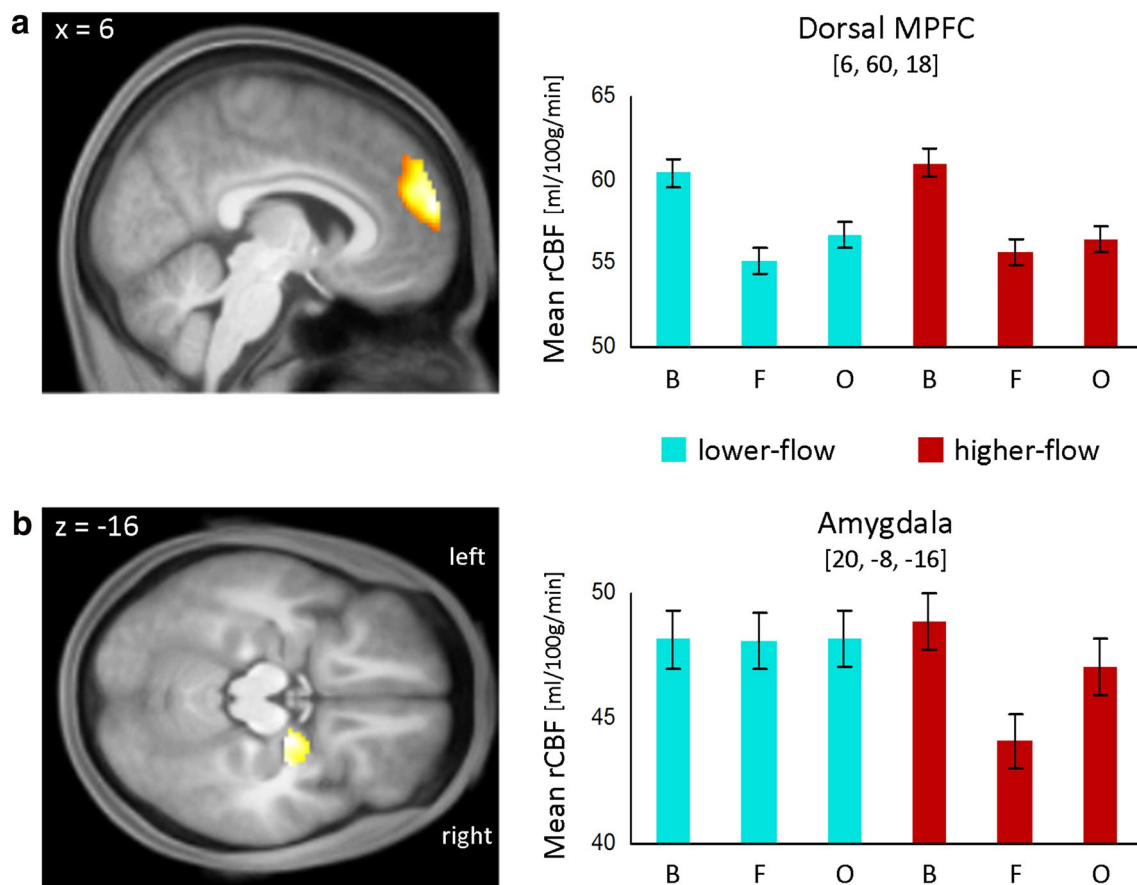


Fig. 2 Perfusion imaging results for the lower-flow (LF) and higher-flow (HF) groups under sham stimulation. **a** The conjoint U-shaped contrast revealed lower regional cerebral blood flow (rCBF) levels during the flow condition (F) than during the boredom (B) and overload (O) conditions in the dorsal medial prefrontal cortex (MPFC) in both groups. **b** Between-group difference (HF>LF) in the right amygdala where the HF group showed a more pronounced U-shaped

response pattern than the LF group. Statistical parametric maps were overlaid on the group averaged T1 image. Coordinates refer to MNI space. The bar chart on the right next to each brain section depicts the mean rCBF levels associated with conditions B, F, and O, averaged across subjects at the peak voxel. Error bars represent standard error of the mean

regressor for this analysis was computed from a cluster where both groups showed a conjoint increase in U-shaped rCBF from sham tDCS to anodal tDCS, exclusively located in ventral anterior MPFC (cluster size: 172 voxels; peak voxel: $[-2, 40, -12]$; z score = 2.52; $p = 0.006$). The regressor was formed by subject-wise subtracting the magnitude of U-shaped rCBF under sham tDCS from that of anodal tDCS. Given this contrast direction, greater positive values describe a more pronounced U-shaped response pattern under anodal tDCS relative to sham. The dependent variable was constructed following the same subtraction logic. Using a regression model as above, group comparisons revealed that now the LF group showed greater positive regression slopes in the subgenual ACC and right amygdala (see Fig. 4b; Table 4) than the HF group. The inverted contrast was associated with greater regression slopes in the dorsal MPFC in the HF group relative to the LF group (see Table 4).

Discussion

In the present study we further explored the neural mechanisms of flow experience using tDCS as a non-invasive method to interfere with cortical activation levels of the MPFC whose differential deactivation during task-based functional imaging has been suggestive to play a causal role in mediating the experience of flow (Ulrich et al. 2014, 2016b). In a sample of 22 male subjects MR-based perfusion imaging implemented as 3D pseudo-continuous arterial spin labeling was used to measure regional cerebral blood flow, while subjects were performing arithmetic tasks of different difficulty levels to implement conditions of boredom, overload, and flow. As predicted by previous studies (see Introduction), baseline dependency in assessing tDCS effects was accounted for by a median split procedure applied to the subjective flow index (Ulrich et al. 2014, 2016b), halving the entire sample into two groups of participants with less

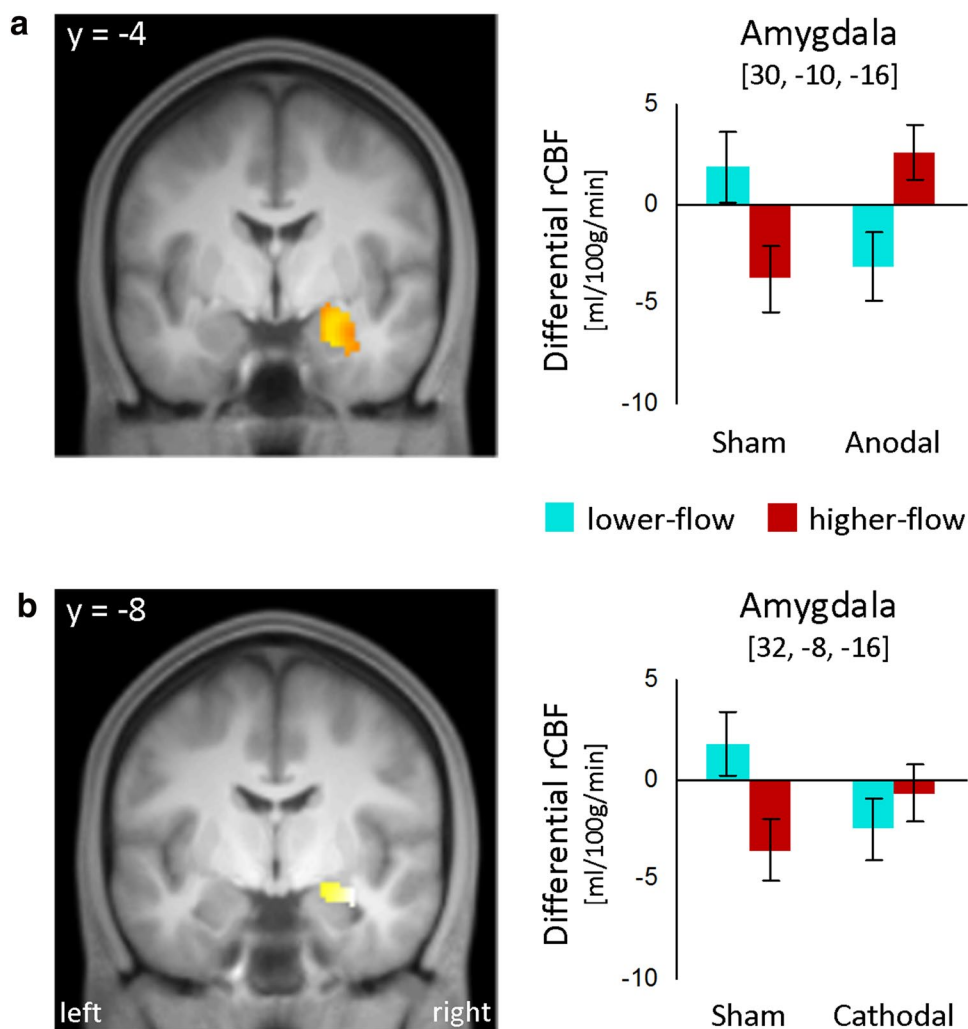
Table 3 Perfusion imaging results for the medial prefrontal cortex and the amygdala as regions of interest

Contrast/anatomical region	Number of voxels	Peak voxel (MNI space)				
		x	y	z	z score	p value
Conjoint U-shaped (B − 2*F + O) activation under sham tDCS in both groups (LF and HF)						
Dorsal medial prefrontal cortex	1216	6	60	18	4.11	0.00002
Dorsal medial prefrontal cortex		−4	58	16	4.01	0.00003
Dorsal medial prefrontal cortex		−4	54	32	3.40	0.00034
Greater degree of U-shaped activation under sham tDCS for the HF group in comparison to the LF group						
Right amygdala	196	20	−8	−16	2.47	0.0068
Right amygdala		32	−8	−18	2.42	0.0079
Greater degree of U-shaped activation under anodal tDCS relative to sham tDCS for the LF group compared with the HF group						
Right amygdala	201	30	−10	−16	3.59	0.0002
Right amygdala		24	−4	−26	2.27	0.0117
Greater degree of U-shaped activation under cathodal tDCS relative to sham tDCS for the LF group compared with the HF group						
Right amygdala	65	32	−8	−16	2.32	0.0102

Significance was assessed at a voxel height threshold $p < 0.050$ and a cluster extent threshold of $k = 30$ voxels

LF lower-flow, HF higher-flow

Fig. 3 Active tDCS treatment effects in the right amygdala. **a** Anodal tDCS treatment vs. sham stimulation. **b** Cathodal tDCS vs. sham stimulation. Bar charts depict the “ $-B + 2 \cdot F - O$ ” difference in regional cerebral blood flow (rCBF) averaged across subjects ($n = 11$ per group). Therefore, the more negative the value, the greater the U-shaped ($B - 2 \cdot F + O$) response pattern. Coordinates refer to MNI space



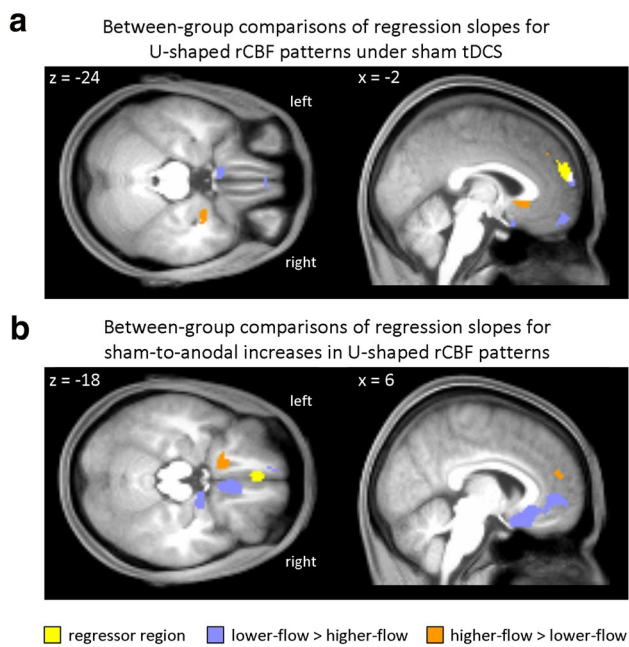


Fig. 4 Results yielded by the regression analyses testing for significant group differences in regression slopes between differential regional cerebral blood flow (rCBF) of a “seed/regressor” region (marked yellow) and other voxels showing the same signal characteristic. **a** Regressor region was the dorsal MPFC cluster where both groups had shown a conjointly significant U-shaped ($B - 2 \cdot F + O$) response pattern during sham stimulation. Relative to the lower-flow (LF) group, the higher-flow (HF) group showed significantly greater positive relationships of U-shaped response patterns in the subgenual anterior cingulate cortex and right amygdala (orange). The opposite effect (LF > HF; cornflower blue) was evident in dorsal and ventral MPFC regions. **b** Regressor region was the anterior ventral MPFC where both groups had shown a conjointly significant increase in the U-shaped response pattern during anodal tDCS relative to sham stimulation. Positive relationships with concordant rCBF responses observed in subgenual anterior cingulate cortex and right amygdala were significantly higher in LF than HF participants (cornflower blue), whereas the opposite direction (HF > LF; orange) was found in the dorsal MPFC and left posterior rectal gyrus. Statistical parametric maps were overlaid on the group averaged T1 image. Coordinates refer to MNI space

(LF) and more (HF) subjective flow experience under sham tDCS. Notably, this division was reflected in the neural data, with the HF group showing a significantly more pronounced U-shaped response pattern in the right amygdala under sham stimulation relative to the LF group. Both groups conjointly demonstrated significant flow-related, relative deactivation of the MPFC at the same location as in our previous studies, corresponding to the observation that both groups had experienced flow. Treatment effects of anodal and cathodal tDCS were then evaluated using two three-way repeated measures analyses of variance with interaction of factors group, treatment and condition in the MPFC and amygdala as regions of interest, with sham stimulation as control condition. Anodal tDCS elicited a significant interaction effect in the

right amygdala where flow-related deactivation was stronger in the LF group corresponding to the LF group’s significant increase in subjective flow experience under this polarity. Conversely, in the HF group, the amygdala’s U-shaped response pattern decreased under anodal tDCS relative to sham, as did this group’s subjective flow index, although not significantly. For cathodal tDCS a rather similar result pattern was observed. Here, the extent of voxels bearing the significant interaction effect was smaller than for anodal tDCS, apparently because the cathodal tDCS-modulatory effect on the U-shaped response patterns appeared smaller in both groups. Changes in the subjective experience of flow were again in correspondence with group-specific changes in the right amygdala’s cerebral blood flow, with an increase in subjective flow experience in the LF group and a decrease in the HF group. However, these changes were not significant in either group.

As expected and informed by previous studies (see “Introduction”) and recent reviews (e.g., Krause and Cohen Kadosh 2014), tDCS effects became evident only when considering baseline differences of flow experience obtained during sham stimulation. We therefore carefully checked whether any external variable could have conditioned the present classification of participants into groups of subjects with less and more flow experience. However, neither ordering of tDCS sessions nor between-group comparisons of performance data, side effects of tDCS stimulation, individual proneness to experience flow, or individual preferences of solving arithmetic tasks did show any indication that would have contributed to present classification of baseline differences. Given that these confounding factors can be ruled out, the classification based on subjective flow experience appears substantial and was also supported by rCBF data demonstrating that a U-shaped response pattern in the MPFC accompanied the flow experiences in both groups and that the plus in flow experience in the HF group was evidently driven by additional, U-shaped activity in the right amygdala.

In a second exploratory analysis we investigated this issue further, particularly with respect to the question of what might have conditioned this group difference in the right amygdala. To this end we computed a regression analysis comparing relationships between a U-shaped response pattern conjointly observed in the MPFC for both groups, and other voxels within the two regions of interest with the same U-shaped response pattern under sham stimulation. This analysis showed clear between-group differences, with the LF group showing positive relationships in ventral anterior parts of the MPFC, while the HF group’s positive relationships were located adjacent to (and partly overlapping with) the “seed” cluster, the right amygdala, and the subgenual part of the ACC. Fully acknowledging that this kind of inter-individual regression

Table 4 Between-group comparisons of regression slopes for U-shaped rCBF patterns under sham tDCS and under anodal (relative to sham) tDCS

Contrast/anatomical region	Number of voxels	Peak voxel (MNI space)				
		x	y	z	z score	p value
Sham stimulation: HF > LF comparison of regression slopes for U-shaped response patterns						
Dorsal MPFC	163	10	46	32	3.01	0.001
Subgenual ACC	76	−2	18	−4	2.25	0.012
Right amygdala	64	34	−2	−24	1.95	0.026
Sham stimulation: LF > HF comparison of regression slopes for U-shaped response patterns						
Ventral MPFC	109	0	56	−20	2.94	0.002
Dorsal MPFC	94	−8	62	14	2.74	0.003
Left posterior rectal gyrus	55	−6	8	−22	2.69	0.004
Anodal—sham stimulation: HF > LF comparison of regression slopes for U-shaped response patterns						
Left posterior lateral orbital gyrus	153	−14	14	−14	3.10	0.001
Dorsal MPFC	89	12	46	24	2.27	0.012
Anodal—sham stimulation: LF > HF comparison of regression slopes for U-shaped response patterns						
Ventral MPFC	980	−10	46	−10	3.29	<0.001
Perigenual ACC		12	38	−8	3.09	0.001
Subgenual ACC		8	26	−12	2.68	0.004
Right amygdala	120	16	−4	−22	2.18	0.014

Significance was assessed at a voxel height threshold $p < 0.050$ and a cluster extent threshold of $k = 30$ voxels

LF lower-flow, HF higher-flow

maps are at best a crude approximation to what is usually known as functional connectivity, results nevertheless suggest that, in participants with higher baseline flow experiences, deactivation of the MPFC might involve a different “route” than that in the LF group, which in the end might have conditioned group differences in the right amygdala. Particularly the subgenual ACC could play an important role in this context given its well-acknowledged role in the functional connection between the MPFC and the amygdala and its role in regulating the amygdala’s activity level (e.g., Zink et al. 2010; Etkin et al. 2011; Holmes et al. 2012; Marusak et al. 2016; Murphy et al. 2016).

In the LF group, treatment effects of both tDCS protocols were associated with significant increase in the right amygdala’s U-shaped response pattern, accompanied by a significant increase in subjective flow experience under anodal tDCS and a close-to-significance increase under cathodal tDCS. This rather linear relationship was not observed in the HF group however. While the U-shaped response pattern significantly reversed under anodal stimulation, it did not under cathodal stimulation. Also, the small numerical declines in the HF group’s subjective flow experience under both tDCS protocols were far from being above chance. The differential between-group result pattern is important insofar as it aligns well with previous reports demonstrating that the so-called “low performers” would usually benefit much more from tDCS than “high performers” who would suffer from tDCS, who usually do not show substantial detrimental effects from tDCS (Hsu et al. 2016).

In another exploratory regression analysis, we compared inter-individual relationships of changes in U-shaped response patterns from sham to anodal stimulation. Interestingly, results revealed almost an entire shift in comparison to the regression results from sham stimulation data. Now, the HF group showed related U-shaped response patterns in the dorsal MPFC, while the LF group’s related response patterns were mainly located in the peri- and subgenual ACC and the right amygdala, thus resembling a possible “signal route” that had originally been observed in HF subjects under sham stimulation.

Clearly, with present perfusion data, our interpretation is to be treated with much caution until replication in a different sample with imaging data permitting greater temporal resolution and therefore much more sophisticated measures of functional (Huskey et al. 2018) and effective connectivity (e.g., Ulrich et al. 2016a). Nevertheless, the between-group sham-to-treatment shift of involvement of the subgenual ACC appears rather promising in raising new hypotheses about how initial coupling of brain regions might change under tDCS stimulation. Further pursuing this idea could also reveal a substantial explanation for the observation that tDCS effects depend on baseline activation, as if the system’s modifiability depended on a neural state imbedded in the inter-individual variation of functionally coupled brain regions.

In line with previous work we did not observe any indication of a polarity effect in its strict sense, that is, opposite effect directions of anodal vs. cathodal tDCS. Instead,

treatment effects were only qualitatively different in their extent when compared against sham stimulation. The existence of polarity effects has already been questioned recently, particularly when cognitive functions are involved (Jacobson et al. 2012; Brückner and Kammer 2017). Comparably to present results, it was concluded that usually anodal stimulation is more likely to interfere with brain activity while cathodal stimulation comes with lower effect sizes, if any. It has been shown (Jacobson et al. 2012; Hsu et al. 2016; Brückner and Kammer 2017) that if anodal stimulation was effective in one particular subject, cathodal stimulation would show the same effect direction in the same subject, although to a smaller extent. At present we do not have any clear explanation for this phenomenon and can only speculate that enhancing cortical excitability of inhibitory interneurons (through anodal tDCS) or diminishing excitability of activating neurons (through cathodal tDCS) could lead to the same net effect, although to various extents.

Limitations

While conditioning of tDCS effects to participants' baseline experience represents a common procedure adopted from previous studies, it appears as rather crude approximation of what is called "initial brain state" thought to interact with neurostimulation, and consideration of which appears mandatory as, otherwise, putative stimulation effects might be underestimated. The relevance of the initial brain state contributing to effect size and effect direction of non-invasive brain stimulation techniques has repeatedly been shown in the field of stimulation studies on motor cortex excitability (for review, see Hurley and Machado 2017). Already in 2004, Lang et al. (2004) and Siebner et al. (2004) showed that the same inhibitory stimulation protocol over the motor cortex had opposite effect directions depending on whether the motor cortex had been preconditioned by an inhibitory or excitatory stimulation protocol. A recent consensus paper (Karabanov et al. 2015) has summarized a series of corresponding studies and also suggested the field to move towards the use of individually adjusted protocols that take relevant neurophysiological state markers into consideration to decrease inter-individual variance, thereby increasing the power of non-invasive transcranial brain stimulation. In that paper it was recommended to measure the dominating oscillation frequency of a target brain network while others have argued to use magnetic resonance spectroscopy to measure the relationship between GABA and glutamate levels as an approximation of the individual balance between cortical excitability and inhibition (Krause and Cohen Kadosh 2014). While both approaches may help to understand the baseline dependency issue of tDCS, it is difficult to say whether they could have been applied in the present work, as both

measures appear best suited to measure an actual brain state at much shorter time lines relative to the present time line during which tDCS was applied. Furthermore, from the background of the regression analyses above we would also suggest to consider differences in brain connectivity as another factor with putative explicative power regarding baseline dependency of tDCS effects. Embedded in an appropriate study design this could also help to resolve an associated issue whether baseline dependency mirrors a state or trait. Treated as a state as in this and other studies, baseline dependency does not necessarily imply stability. However, if, for instance, functional connectivity could be seen as a trait marker conditioning tDCS effects, this would certainly increase explicative power and motivate further research investigating reasons for inter-individual trait differences.

Another limitation of the actual study has to do with selection of males only which was motivated by recent data demonstrating hormonal variations during the menstrual cycle can affect magnitude and local distribution of cerebral hemodynamics (e.g., Dietrich et al. 2001; Hausmann et al. 2002; Fernández et al. 2003; Abler et al. 2013; Pletzer et al. 2013) and of tDCS effects (Krause and Cohen Kadosh 2014). Insofar the interpretation of present results remains restricted to men only.

Finally, and particularly related to the ongoing debate about reliability and replicability of neuroimaging results, a greater than usual false positive risk was taken. However, actual recommendations (Eklund et al. 2016; Nichols et al. 2017) had been derived from typical MR-based BOLD imaging data which is known to come with much greater contrast-to-noise ratio than MR-based perfusion imaging (Yang et al. 2005; Liu and Brown 2007; Viviani et al. 2011; Wang et al. 2011; Detre et al. 2012; Donahue and Jezzard 2013). Additionally, it should be considered that inference of tDCS effects was always computed in the presence of rather specific interaction contrasts where a condition difference between the flow and its control conditions was always present. Therefore, whether present acceptance of a greater than usual error probability was a failure or not, can only be explored in a replication study using BOLD instead of perfusion imaging, also because of the much greater temporal resolution of BOLD imaging compared to perfusion imaging, particularly in combination with simultaneous multislice imaging (Barth et al. 2016). In this context, however, it also remains an open issue whether rather gradual inter-individual differences in subjectively experiencing flow observed in healthy young controls do provide a well enough basis for expecting effect sizes that would signal greater significance than present. Perhaps some greater focus on subjects more different in their propensity to experience flow may provide greater contrast-to-noise ratios in evaluating tDCS effects that could help in bolstering their reliability.

Conclusions

In the present study we have tested whether tDCS over the medial prefrontal cortex can be used to further investigate the relevance of this brain area in mediating the experience of flow, which has been suggested by our previous studies (Ulrich et al. 2014, 2016b). TDCS effects depended on baseline flow experiences during sham stimulation, appeared downstream in the right amygdala, were greater for anodal than cathodal stimulation, did not show any indications of opposite effect directions as a function of polarity, and subjects with less subjective flow experience (LF) during sham profited more from active tDCS than this treatment was detrimental to subjects with initially greater flow experience (HF). Additional analyses suggested that tDCS over the forehead during the flow paradigm may have entered the brain via a ventral aspect of the anterior MPFC where anodal tDCS appeared to have increased the flow condition-specific decrease in neural activation in both LF and HF participants. This change in activation was related to group-specific changes in activations of the subgenual anterior cingulate cortex and the right amygdala, suggesting that the typically reported baseline dependency of tDCS might be grounded in individual variations of the path the tDCS-modulated signal travels along. However, despite its various advantages in the presence of low-fluctuating stimulation paradigms, present MR-based perfusion imaging has too low a temporal resolution to draw any firm conclusions. This, however, should become possible once the present experiment is replicated in a larger sample employing simultaneous multislice BOLD imaging techniques in combination with more sophisticated models of functional and effective connectivity. It is not unreasonable to assume that baseline dependency of tDCS effects could be explained in greater portions by individual variations in brain connectivity, which would lend this phenomenon greater empirical support. It would also reflect a crucial feature to better understand the generation of tDCS effects which could then be exploited to further fostering our knowledge about the neural mechanisms of flow experience.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

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